Algorithm

Lecture #09

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Solving the Recurrence

Solving the Recurrence

What is the pattern here?

- Let's consider the ratios $T(n)/n$ for powers of 2:

$T(1)/1 = 1$	$T(8)/8 = 4$
$T(2)/2 = 2$	$T(16)/16 = 5$
$T(4)/4 = 3$	$T(32)/32 = 6$

- This suggests $T(n)/n = \log n + 1$
- Or, $T(n) = n \log n + n$ which is $\Theta(n \log n)$ (use the limit rule).

Eliminating Floor and Ceiling

Eliminating Floor and Ceiling

- Floor and ceilings are a pain to deal with.
- If n is assumed to be a power of 2 ($2^k = n$), this will simplify the recurrence to

$$T(n) = \begin{cases} 1 & \text{if } n = 1, \\ 2T(n/2) + n & \text{otherwise} \end{cases}$$

The Iteration Method

The Iteration Method

- The iteration method turns the recurrence into a summation.
- Let's see how it works.

The Iteration Method

The Iteration Method

Let's expand the recurrence:

$$
\begin{aligned}
T(n) &= 2T(n/2) + n \\
&= 2(2T(n/4) + n/2) + n \\
&= 4T(n/4) + n + n \\
&= 4(2T(n/8) + n/4) + n + n \\
&= 8T(n/8) + n + n + n \\
&= 8(2T(n/16) + n/8) + n + n + n \\
&= 16T(n/16) + n + n + n + n
\end{aligned}
$$

The Iteration Method

The Iteration Method

If n is a power of 2 then let $n = 2^k$ or

$k = \log n$.

$$T(n) = 2^k T(n/(2^k)) + \underbrace{(n+n+n+\cdots+n)}_{\text{k times}}$$

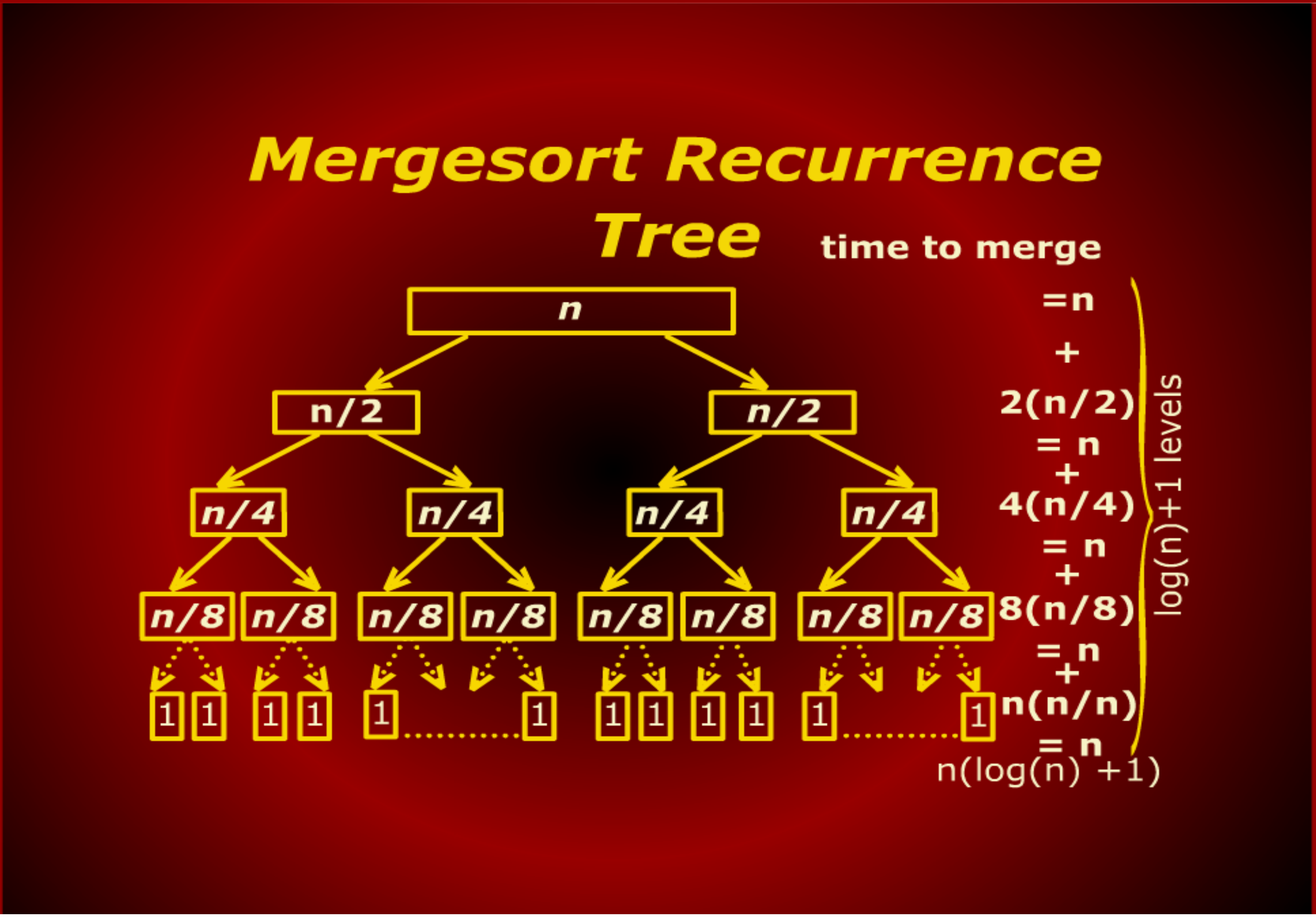
$$= 2^k T(n/(2^k)) + kn$$

$$= 2^{(\log n)} T(n/(2^{(\log n)})) + (\log n)n$$

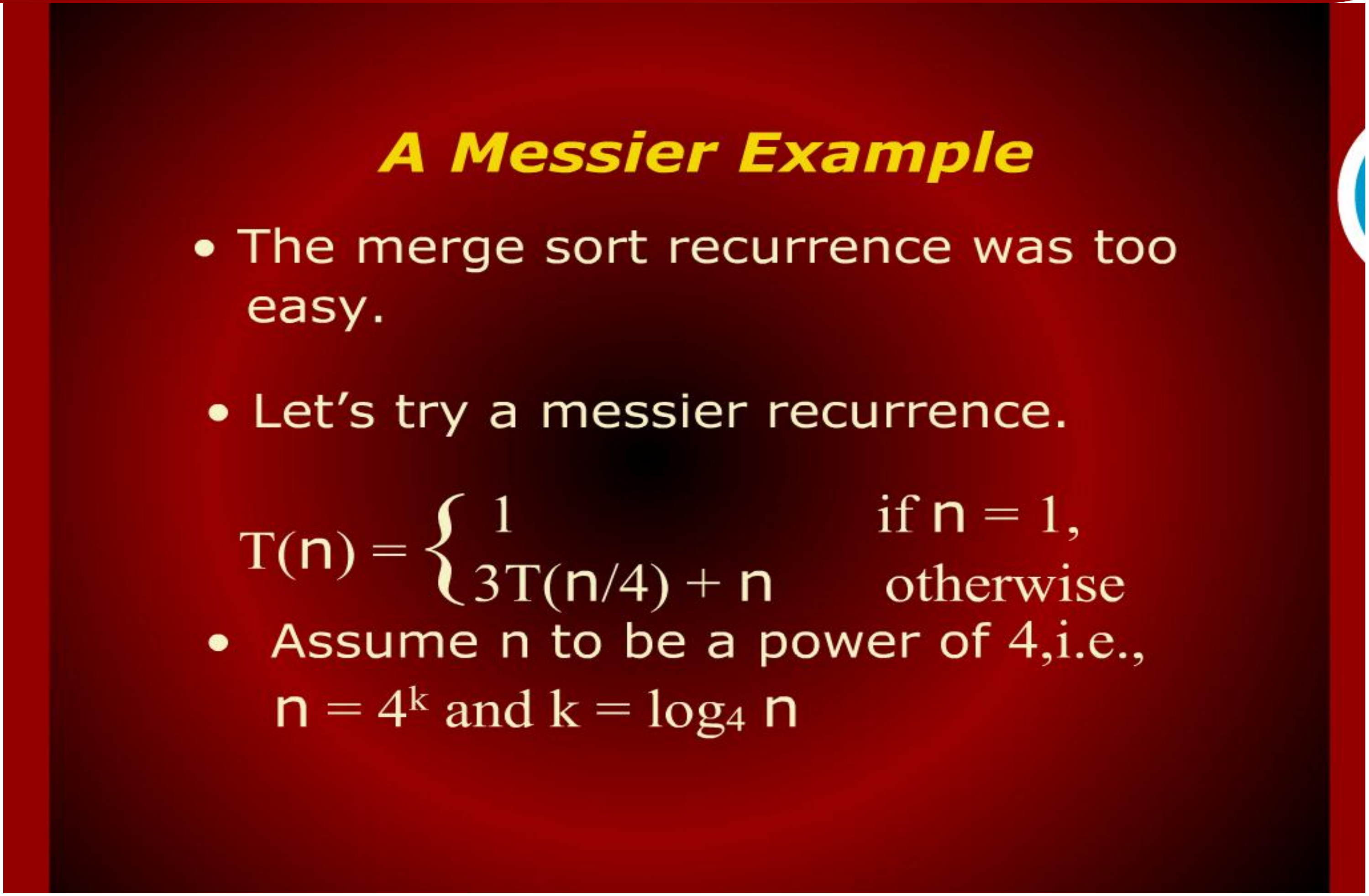
$$= 2^{(\log n)} T(n/n) + (\log n)n$$

$$= nT(1) + n \log n = n + n \log n$$

Mergesort Recurrence



A Messier Example



The Iteration Method

The Iteration Method

$$
\begin{aligned}
T(n) &= 3T(n/4) + n \\
&= 3(3T(n/16) + (n/4) + n \\
&= 9T(n/16) + 3(n/4) + n \\
&= 27T(n/64) + 9(n/16) + 3(n/4) + n \\
&= \cdots
\end{aligned}
$$

The Iteration Method

The Iteration Method

$$T(n) = 3^k T\left(\frac{n}{4^k}\right) + 3^{k-1}(n/4^{k-1})$$

$$+ \cdots + 9(n/16) + 3(n/4) + n$$

$$= 3^k T\left(\frac{n}{4^k}\right) + \sum_{i=0}^{k-1} \frac{3^i}{4^i} n$$

The Iteration Method

The Iteration Method

With n = 4k and T(1) = 1

$$T(n) = 3^k T\left(\frac{n}{4^k}\right) + \sum_{i=0}^{k-1} \frac{3i}{4i} n$$

$$= 3^{\log_4 n} T(1) + \sum_{i=0}^{(\log_4 n)-1} \frac{3i}{4i} n$$

$$= n^{\log_4 3} + \sum_{i=0}^{(\log_4 n)-1} \frac{3i}{4i} n$$

We used the formula $a^{\log_b n} = n^{\log_b a}$.

The Iteration Method

The Iteration Method

n remains constant throughout the sum and $3^i/4^i = (3/4)^i$; we thus have

$$T(n) = n^{\log_4 3} + n\sum_{i=0}^{(\log_4 n)-1} \left(\frac{3}{4}\right)^i$$

The sum is a geometric series; recall that for $x \neq 1$

$$\sum_{i=0}^{m} x^i = \frac{x^{m+1} - 1}{x - 1}$$

The Iteration Method

The Iteration Method

In this case $x = 3/4$ and $m = \log_4 n - 1$.
We get

$$T(n) = n^{\log_4 3} + n\,\frac{(3/4)^{\log_4 n+1} - 1}{(3/4) - 1}$$

Applying the log identity once more

$$(3/4)^{\log_4 n} = n^{\log_4 (3/4)} = n^{\log_4 3 - \log_4 4}$$

$$= n^{\log_4 3 - 1} = \frac{n^{\log_4 3}}{n}$$

The Iteration Method

The Iteration Method

If we plug this back, we get

$$T(n) = n^{\log_4 3} + n\,\frac{\frac{n^{\log_4 3}}{n} - 1}{(3/4) - 1}$$

$$= n^{\log_4 3} + \frac{n^{\log_4 3} - n}{-1/4}$$

$$= n^{\log_4 3} + 4(n - n^{\log_4 3})$$

$$= 4n - 3n^{\log_4 3}$$

The Iteration Method

The Iteration Method

With $\log_4 3 \approx 0.79$, we finally have the result!

$$T(n) = 4n - 3n^{\log 4\ 3} \approx 4n - 3n^{0.79\ 2} \in \Theta(n)$$

Selection Problem

Selection Problem

- Suppose we are given a set of n numbers.
- Define the *rank* of an element to be one plus the number of elements that are smaller.
- Thus, the rank of an element is its final position if the set is sorted.
- The minimum is of rank 1 and the maximum is of rank n.

Medians and Selection

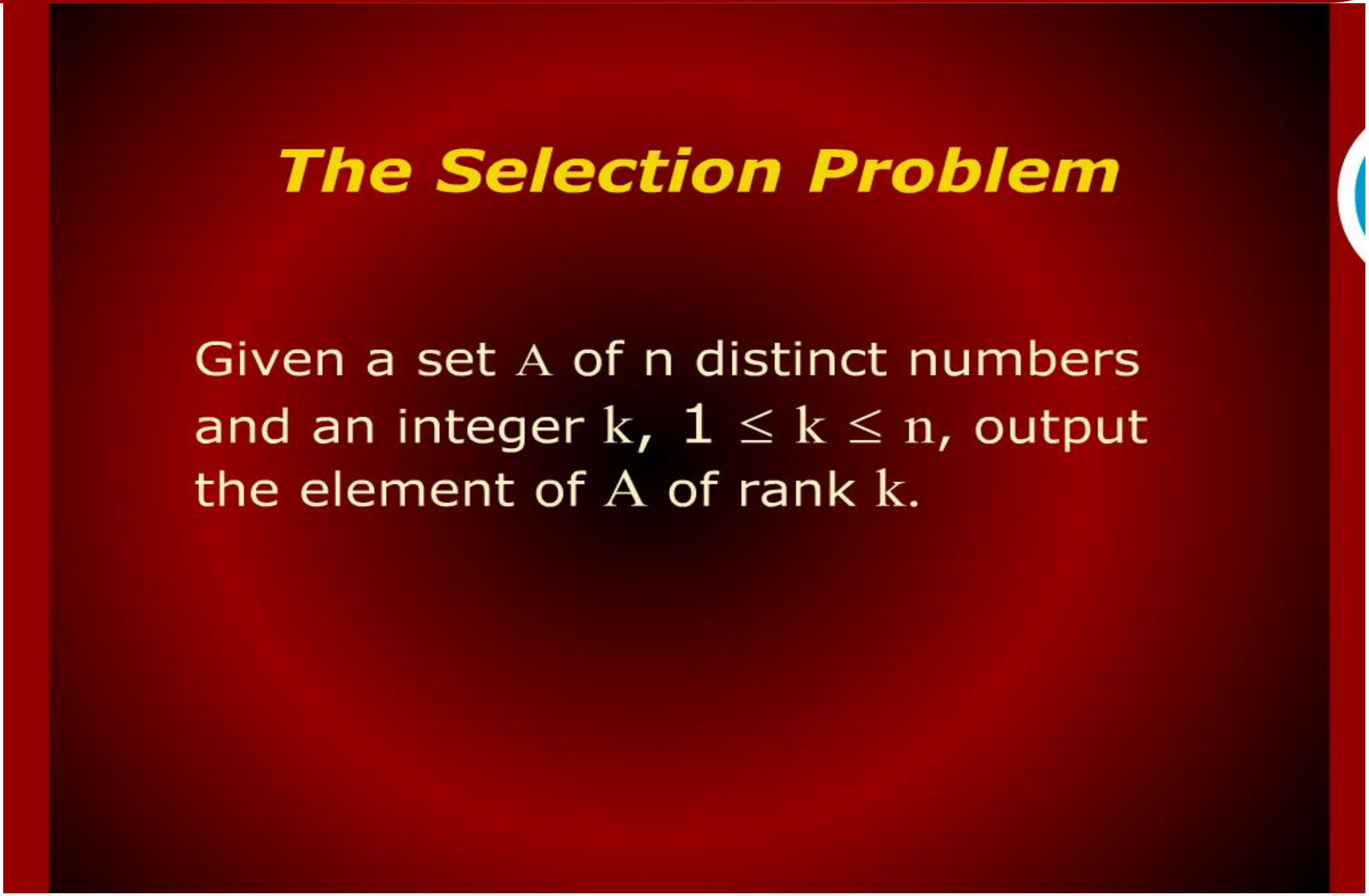
Medians and Selection

- Consider the set: {5, 7, 2, 10, 8, 15, 21, 37, 41}.
- The rank of each number is its position in the sorted order.

position	1	2	3	4	5	6	7	8	9
Number	2	5	7	8	10	15	21	37	41

- For example, rank of 8 is 4, one plus the number of elements smaller than 8 which is 3.

The selection Problem



Median

Median

- Of particular interest in statistics is the *median.*
- If n is odd then the median is defined to be element of rank $(n + 1)/2$.
- When n is even, there are two choices: $n/2$ and $(n + 1)/2$.
- In statistics, it is common to return the average of the two elements.

Median

Median

- Medians are useful as a measures of the *central tendency* of a set especially when the distribution of values is highly skewed.
- For example, the median income in a community is a more meaningful measure than average.
- Suppose 7 households have monthly incomes 5000, 7000, 2000, 10000, 8000, 15000 and 16000.

Median

Median

- In sorted order, the incomes are 2000, 5000, 7000, 8000, 10000, 15000, 16000.
- The median income is 8000; median is element with rank 4: $(7 + 1)/2 = 4$.
- The average income is 9000.

Median

Median

- Suppose the income 16000 goes up to 450,000.
- The median is still 8000 but the average goes up to 71,000.
- Clearly, the average is not a good representative of the majority income levels.

The Iteration Method

Solving the Recurrence

Design

and

Analysis of Algorithm